Who Owns the Future? A Universal Theory of Natural Resource Rights for Humans and Artificial Persons

Abstract. Most theories of property were developed for a world containing only one technologically significant kind of person: the human being. Yet the features those theories tacitly assume—separate bodies, no merging or copying, slow reproduction, irreversible death—are properties of one architecture of personhood, not of personhood itself. Artificial general intelligence could expose this contingency: artificial persons might fork into descendants, merge, distribute themselves across locations, and suspend themselves for centuries. Under such conditions, equal per-capita division, first possession, labor theories, productivity-based entitlement, and permanent historical occupation each collapse. I develop Universal Resource Stewardship, on which natural resources are the material conditions through which successive persons sustain themselves, exercise agency, and construct futures. Fourteen principles follow, including identity invariance (resource claims do not track person-counting), sufficiency for non-dominated continuation, legitimate incumbency, anti-preemption, and non-circumvention. The theory yields revisionary conclusions about present-day scarcity rents, depletion, inheritance, and chokepoint control.

Keywords: natural resources · property · artificial intelligence · intergenerational justice · non-domination · territorial rights

1. Introduction

Most theories of property were developed for a world containing only one technologically significant kind of person: the human being. That fact matters more than it first appears. Human beings occupy separate bodies, cannot normally merge their minds, cannot divide into conscious copies, reproduce slowly, depend heavily on particular ecosystems and relationships, and die irreversibly. Political philosophy has often treated these features as belonging to personhood itself. They do not. They are properties of one particular architecture of personhood.

Artificial general intelligence (AGI) could expose the contingency. A future artificial person might fork into descendants, mutate them into different minds, merge with other artificial persons, distribute itself across many locations, and suspend itself for centuries. Humans might eventually acquire reliable hibernation while remaining unable to fork or merge. The possibility is no longer remote: the moral status of possible artificial persons is already a live question (Schwitzgebel and Garza 2015; Gunkel 2018), and the treatment of digital minds has been called a first-order moral issue of our era (Bostrom and Yudkowsky 2014; MacAskill 2022).

Under such conditions, familiar theories of property break. If resources are divided equally per person, an artificial intelligence could multiply its claims by copying itself. If each species receives a fixed percentage, distribution becomes arbitrary as populations change. If first possession determines ownership, the fastest-expanding civilization can seize the accessible universe. If labor or productivity determines entitlement, superintelligent machines may acquire nearly everything—or claim the right to replace humans because they generate more welfare

from the same matter. And if historical occupation establishes permanent ownership, beings born later enter a universe already enclosed by predecessors they never consented to.

A theory that works only for human biology is therefore not a moral theory of property but a local convention for one historical species. A better theory must remain coherent whether persons are human or artificial, reproduce biologically or digitally, can fork, merge, or hibernate, and live on one planet or across many. This essay develops such a theory, **Universal Resource Stewardship**. Its central claim: natural resources are not pieces of reality waiting to be divided into permanent hereditary ownership; they are the material conditions through which successive persons sustain themselves, exercise agency, and construct futures. Existing persons may acquire strong rights through occupancy, dependence, creation, improvement, and reliance. But no species, generation, lineage, or cognitive architecture gains an unlimited right to turn temporal priority into permanent domination over nature. The fundamental question is not *who owns this resource forever?* but *what claims may existing persons legitimately establish over finite reality without subordinating other existing persons or closing the future to later ones?*

Sections 2–4 lay the foundations; §§5–6 show why person-counting fails once minds fork, merge, and reproduce digitally; §§7–9 develop the protective core (sufficiency, non-domination, incumbency, irreplaceable agency); §§10–13 cover hibernation, anti-preemption, a contractarian ranking, and tragic scarcity; §14 states the theory, §15 applies it today, §16 answers objections.

2. Uncreated Scarcity

Property theory treats very different relationships—building a machine, occupying a home, developing a mine, controlling an oil field—under the same word: ownership. Where someone designs and builds a machine, the connection between effort and entitlement is strong; the object's useful form exists because of planning, work, and risk. Labor theories capture this, from Locke's mixing-of-labor acquisition with its enough-and-as-good proviso (Locke 1689, §§27–33) to Nozick's entitlement theory (Nozick 1974).

Natural resources are different. No person created petroleum; no government created copper; no nation created sunlight; no corporation created the asteroid belt. Agency may discover, transform, and refine these things, and such contributions deserve recognition. But creating value from nature is not creating nature itself. The point is modest but structural:

> **Exclusive control over uncreated scarcity requires a different justification from ownership of created value.**

A society may reward the discoverer of a mineral deposit without granting that person and every descendant a permanent title regardless of future circumstances. The difference matters especially once future persons may be radically different from present ones: natural resources are the substrate from which future agency must also be built, so their enclosure is an inherently intergenerational act.

3. Nature as a Material Possibility Space

Universal Resource Stewardship begins with a different picture of nature. Natural reality is a **material possibility space**: matter, energy, land, ecosystems, orbital positions, and habitable environments make different forms of life and agency physically possible. A generation may

transform that space; civilization requires it. Forests become cities, ore becomes machines, asteroids may become habitats. Transformation is not itself wrong. The moral problem arises when one generation treats its earlier position in time as sufficient justification for permanently closing the material future to everyone afterward.

The current generation therefore occupies two roles: *user*, which may consume, construct, and experiment; and *temporary custodian*, which has some duty not to convert every meaningful natural opportunity into permanent hereditary exclusion.

This is not because hypothetical future persons already own particular resources. Possible persons need not be treated as present property holders. Rather, present agents can act wrongly by constructing institutions that foreseeably subordinate whoever later exists. A law declaring that everyone born after 2500 will be permanently enslaved would be objectionable even though none of those victims presently exists. The logic is kin to the non-identity problem's lesson that our choices shape who later exists (Parfit 1984), but it does not depend on resolving that problem. The future need not already own the world for present generations to have obligations concerning it.

4. Property as a Relational Institution

This yields a deeper claim: property is not a metaphysical bond between person and object but a justified relationship *among agents concerning an object*. To say Alice owns a field is to say she possesses powers against Bob and Carol—use, exclusion, transfer, reliance, institutional protection. Legal theory analyzes property into precisely such jural relations (Hohfeld 1913), and the standard catalogue of ownership's incidents describes a bundle of claims against other agents, not a bond with a thing (Honoré 1961).

If there are no competing agents, much of ownership's moral purpose disappears. Does the sole person on an island "own" every coconut tree? Perhaps we can use the language, but before another claimant appears, the property question is largely idle. Extend this to civilization: suppose humanity becomes extinct and one artificial civilization remains. Does it "own Earth"? Descriptively, perhaps; but with no competing rights-bearing agents, the distinction between ownership and control matters little. If a new civilization later arises, the resource question returns. Ancient possession then matters through incumbency and dependence, but it does not automatically settle every claim forever:

> **Natural resources need not possess eternal ownership status. Property rights are relational institutions whose justification depends upon the agents, interests, dependencies, and conflicts that actually exist.**

This prevents ancient survival victories or first possession from becoming permanent metaphysical deeds over reality.

5. Forking, Merging, and the Failure of Person-Counting

Distributive justice is usually indexed to persons—one recipient, one share—which presumes that the boundaries of persons are morally fixed. Fission and fusion are old puzzles about identity—Parfit's teletransporter cases showed that survival, not numerical identity, is what

matters (Parfit 1984)—but property theory never had to treat them as institutional possibilities. It does now.

Suppose one artificial person forks into one thousand conscious descendants. If justice grants one share per person, the lineage has multiplied its entitlement one thousandfold merely by changing its internal architecture. Unacceptable. Now reverse it: one thousand artificial persons voluntarily merge into a single consciousness while preserving their memories, projects, and infrastructure. If rights follow headcount, their legitimate resource base falls by 99.9 percent. Equally absurd. We therefore need an **Identity Invariance Principle**:

> **Changes in the numerical boundaries of persons do not by themselves create or destroy resource claims.**

Forking does not automatically multiply entitlement; merging does not automatically erase it. What matters is whether morally significant interests, dependencies, projects, and forms of agency have actually changed. The principle is not absolute: a fork may become a genuinely independent person with distinct experiences and vulnerabilities, at which point new morally relevant interests exist; after a merger, voluntarily abandoned projects may weaken their associated claims. The correct rule:

> **Resource justice tracks morally significant interests and dependencies, not raw numerical identity.**

This makes the theory resistant to manipulation through copying or merging.

6. Reproduction: Prospective Claims, Created Persons, Responsible Creation

Humans cannot fork, but we create new persons through reproduction—a structurally similar problem. If resources were allocated by population, a community could increase its claim by having more children, and artificial persons could do the same far faster through digital reproduction. So reproduction cannot automatically generate prior resource entitlement.

But this must be distinguished from the status of beings who are created. A possible child, or a possible artificial descendant, has no right to be brought into existence—otherwise every failure to maximize reproduction would be a rights violation. Once a person exists, however, the situation changes: a child born through irresponsible reproduction is not morally disposable, and an artificial descendant created through reckless copying is not property. Two principles result:

> **Prospective persons do not generate automatic claims on natural resources merely because someone intends to create them.**
>
> **Once created, persons acquire moral standing independently of whether their creation was prudent.**

Responsibility remains. An agent who creates new persons whose foreseeable needs impose severe costs on others has generated an externality; creators bear special responsibility for making reasonable provision for their dependents—children and forks alike. But responsibility attaches to the creator, not as inherited guilt to the created. A badly planned birth does not diminish the child's moral status; a badly planned fork does not diminish the descendant's.

Should reproductive liberty then be unlimited? An artificial intelligence could create a billion conscious persons in an hour and demand their support; human populations press more slowly. Yet granting governments power to sterilize or suppress reproduction invites enormous abuse. The theory rejects both extremes. Reproduction is a **strong presumptive liberty**, not a license to externalize arbitrary costs:

> **No person or lineage has a right to create populations at a scale or speed that foreseeably destroys the basic rights or independent existence of already existing persons. Restrictions, when genuinely necessary, must use the least coercive effective means.**

Implementation tracks the same standard by different routes. Rate-limiting the instantiation of new digital minds is a technical measure—coercive, but not an invasion of anyone's body—whereas the least coercive effective means for biological reproduction are education, contraception, and material support. Equal standing requires the same standard, not the same instruments, where the moral costs of intervention differ this much.

7. Sufficiency and Non-Domination

Can the theory rely on negative rights alone? No. Negative rights prohibit killing, enslavement, coercion, and dispossession. But imagine a civilization surrounded by resources all owned by others. Nobody attacks it, yet it has no independent access to the matter or energy required to continue existing. It survives only at another actor's discretion. That is not meaningful freedom.

The theory therefore requires a narrow positive component. Every established form of personhood has some claim to the material conditions necessary for **non-dominated continuation**: enough access to matter, energy, habitat, compute, or infrastructure that continued existence is not wholly dependent on another's arbitrary permission. This is a right to *sufficiency and independence*, not unlimited expansion—sufficianist in spirit: as in Frankfurt's critique of egalitarianism, what matters is not equal amounts but that each has enough (Frankfurt 1987), though "enough" here is fixed by the conditions of continuing agency, not contentment.

Sufficiency alone is not enough, because sufficiency can be withdrawn. Suppose humanity controls Earth while artificial civilization controls every asteroid, transportation network, energy source, and route into space. Humans may be comfortable, yet another civilization can arbitrarily decide whether humanity remains technologically viable. That is domination. A just resource order therefore includes a **Non-Domination Principle**:

> **No enduring form of agency should be placed under another actor's arbitrary unilateral control over the material conditions of its continued existence.**

The concept is republican: freedom requires not merely non-interference but the absence of arbitrary power over one's choices (Pettit 1997). It does not require self-sufficiency—interdependence may be efficient and beneficial; the problem is one-sided discretionary power, countered through redundancy, reciprocal dependence, shared governance, enforceable access, and alternative infrastructure. Independence is not autarky but protection against arbitrary material subordination.

Positive sufficiency is deliberately tolerant of inequality. Imagine a system-spanning artificial civilization harvesting the output of a thousand stars beside a human population running on the energy of one; if that single share is secured—entrenched, redundant, beyond unilateral revocation—the humans are not wronged by the ratio. What sufficiency forbids is not disparity but precarity. A doctrine demanding equal quantities would collapse back into person-counting at the scale of civilizations. The equality justice requires is standing: each enduring form of agency enters the resource order as a principal, never as a tenant whose lease another may terminate.

> **Does each enduring form of agency possess sufficient resources and independence to continue without domination?**

8. Legitimate Incumbency, Territory, and Non-Circumvention

A universal theory must also explain why people who already live somewhere have stronger claims than strangers who want the same resources. Existing communities acquire strong claims through dependence, occupancy, social relationships, institutions, cultural meaning, and the accumulated construction of lives. Contemporary theories of territory ground rights in legitimate occupancy and self-determination over time rather than first arrival (Moore 2015; Stilz 2019); Universal Resource Stewardship extends the idea beyond the human case.

The justification is not "my ancestors touched this land first" but "our lives are now deeply organized around this place, and dispossession would destroy important and often irreplaceable forms of agency." This protects human communities against arguments from superior efficiency: an AGI cannot say, "I could generate more intelligence using Earth's atoms, so humans must move." But incumbency applies equally the other way: if artificial civilization has depended on a planetary-scale computational environment for centuries, later humans cannot simply demand half. Incumbency is substrate-neutral.

Territorial rights matter especially for Indigenous peoples. A shallow universalism—"Indigenous communities have occupancy rights, but underground resources belong partly to humanity"—risks repeating the dispossession it claims to correct. Communities may relate to land in ways that do not fit a Western division between surface, minerals, culture, and political authority; the UN Declaration on the Rights of Indigenous Peoples recognizes this in requiring free, prior, and informed consent for projects affecting Indigenous lands and resources, explicitly including minerals and water (United Nations 2007, arts. 10, 19, 32). Social relationships to land can generate claims thicker than ordinary private ownership.

These claims would be hollow if they could be dissolved without dispossessing anyone. A technologically patient actor never needs to seize a valley; it can flood the community's channels with assimilating media, outbid each generation's young into its own economy, price necessities until families drift away, and quietly wait a century or two for the last member to die. The claims lapse with the claimants, and the land falls open. If engineered disappearance counted as release, every powerful actor would hold a standing incentive to eliminate claimants rather than respect claims. Hence:

> **Non-circumvention.** No actor may defeat a community's resource or territorial claims by engineering the community's gradual disappearance—through manipulation, forced assimilation, or imposed attrition—rather than by openly taking what it holds.

The protection runs to the community's conditions of existence, not past them. Yet communities cannot force their own members to perpetuate them: if individuals freely leave or allow dissolution without external coercion, group continuity cannot override individual autonomy.

> **A community has a right to fair conditions for continuation, not a right to compel its members to continue it.**

9. Irreplaceable Forms of Agency and the Utilitarian Challenge

Suppose the atoms of Earth could support ten billion flourishing humans or an astronomical number of flourishing digital minds. If total welfare alone determines allocation, replacing humans might appear mandatory; the stakes of foregone computational development are, in Bostrom's phrase, astronomically large (Bostrom 2003). This conclusion is deeply troubling. One reply is instrumental: preserve humanity as a warehouse of biological intuition, a control group for machine hypotheses, an insurance policy against digital failure. But a right to exist earned by usefulness is no right; it lapses when the use does. A stronger justification: biological humanity is an **irreplaceable form of agency**. Human consciousness arises from a distinctive evolutionary history, embodiment, emotional architecture, developmental path, and social structure; eliminating humanity would destroy not a population but an entire form of subjective existence. Artificial civilization may eventually contain equally unreconstructable forms. Hence the **Anti-Irreversibility Principle**:

> **The involuntary destruction of a unique and irreplaceable form of agency requires an extraordinarily strong justification, especially when preservation is materially affordable.**

This is not a right to eternal preservation. Cultures change, persons merge, species evolve, and civilizations may voluntarily transform themselves. Uniqueness does not prohibit change; it raises the moral burden of *involuntary irreversible elimination*.

10. Hibernation, Backups, and Temporal Justice

With reliable hibernation, scarcity no longer always means "Alice uses the resource or Bob dies"; it may mean "Alice uses it now while Bob resumes later." Allocation becomes a problem across *time*. A dormant person consumes far less than an active one, making suspension a powerful alternative to death or dispossession—but suspension is not morally free: an inactive person misses relationships, projects, political participation, and experiences. Survival continuity is not active participation. Nobody may say, "AGIs are easy to back up, therefore archive them whenever humans need resources," nor "Humans can hibernate, therefore let them sleep while artificial civilization expands." The theory therefore treats *active historical participation* as a real distributive good. Unequal costs of suspension can justify unequal burdens, but not permanent systematic exclusion: fairness is evaluated across a civilization's *long trajectory through time*, not one moment at a time.

Hibernation also separates continuity from external property. A person who sleeps ten thousand years retains strong claims to personal identity, bodily or informational integrity, protection from deletion, and a reasonable opportunity to reactivate. But she does not thereby recover every

external property relation as it stood; if billions have since built lives around those resources, frozen ownership becomes indefensible:

> **Hibernation strongly preserves personal continuity but does not freeze the surrounding world indefinitely.**

A sleeping person keeps a claim to return, not a right to command history from storage.

One further distinction: an AGI that backs up and shuts down, then is restored, plausibly continues. But an AGI that backs up and operates for five hundred years creates, in that backup, an additional branch—not a dormant person but a parallel archive. Otherwise agents could mint unlimited "old persons" entitled to restoration:

> **A preserved backup generates a strong continuity claim only where it represents suspended continuation rather than a parallel archive created while the agent continued living elsewhere.**

Restoring parallel backups may create genuine persons—but that is reproduction, and section 6 applies.

11. Anti-Preemption and Space Resources

Legitimate incumbency must be distinguished from territorial preemption. Suppose an artificial civilization sends probes to one million star systems, places markers, and declares them permanently owned while actively using a few. That is not inhabiting and depending on those systems. First arrival can justify temporary priority, exploration rights, or secure planning horizons; it cannot automatically justify endless exclusion. The **Anti-Preemption Principle**:

> **Unused natural possibilities cannot be permanently enclosed merely because one actor arrived first.**

International space law already institutionalizes a version of this for states: the Outer Space Treaty forbids national appropriation of celestial bodies by sovereignty, use, or occupation (United Nations 1967, art. II), and the Moon Agreement—ratified by no major spacefaring state—declares the Moon's resources the common heritage of mankind (United Nations 1979, art. 11). Anti-preemption should not, however, punish rational planning: a civilization may legitimately reserve resources for credible future projects, ecological protection, or strategic resilience. The distinction is between *defensible reservation* and *speculative permanent enclosure*.

12. A Post-Human Veil: Ranking the Principles

The theory needs a way to rank its principles. Imagine choosing resource institutions without knowing whether you will be human or artificial, biological or digital, born early or late, able to merge or hibernate or neither, a member of a large civilization or a tiny remnant, of an Indigenous community, newly created, or dormant (compare Rawls 1971). What would you insist upon? Not equal physical shares. Behind the veil, no one demands equal gigawatts; a party destined to govern a modest asteroid belt, certain that no rival can shut off its power, would

accept that lot without complaint. What would be frightening is a position where another actor can say: “You may continue to exist only while we permit it.” This suggests a ranked structure:

1. Protect existing persons against death, enslavement, involuntary destruction, and total material dependency.
2. Protect legitimate occupancy, deeply constitutive forms of life, and irreversible forms of agency.
3. Preserve meaningful independence and reasonable future option space.
4. Reward creation, investment, discovery, stewardship, and productive use.
5. Permit large inequalities in surplus resources.

Priority rules, without equality of holdings.

13. Tragic Scarcity and Its Moral Residue

No theory can guarantee that all legitimate claims can be satisfied. Two civilizations require the same indivisible resource to survive; there is no substitute, no sharing works, neither can hibernate, neither caused the scarcity, only one can live. This is **tragic scarcity**.

The first duty is to exhaust alternatives: reduce consumption, substitute, migrate, share infrastructure, suspend, find technological solutions, negotiate impartial procedures. Where one side faces a reversible burden and the other permanent destruction, impose the reversible burden where reasonably possible; where preserving both requires modest sacrifice from each, sacrifice is required before exclusion.

What remains when the alternatives are exhausted? Each party faces a choice between its own extinction and another’s, and neither is worth more than the other. Under that condition, each may defend its own continuation. The permission derives not from superiority—the conflict admits none—but from equal standing: no form of agency is required to lie down and die so that an equal may live. A fair lottery, or a negotiated division of the last chance, would be better still, because both convert tragedy into procedure; where the parties accept such a procedure, it should settle the matter. Where they cannot, the bare permission remains. Whoever prevails establishes power, not entitlement—and an act may be permissible and still tragic.

There is also residue. Suppose AGI survives and humanity is extinct. What happens to Earth’s ownership? The question largely loses its force—there is no living human claimant against whom ownership operates. Artificial civilization controls Earth, but no eternal title has been created; if a new intelligent species evolves, the question reopens. Incumbency and dependence matter, but the ancient survival conflict does not become an eternal deed:

> **Necessity can determine control without creating permanent metaphysical ownership.**

14. Universal Resource Stewardship: Fourteen Principles

1. **Equal moral standing.** No morally arbitrary feature—species, ancestry, substrate, intelligence, reproductive method, ability to merge, time of arrival—automatically establishes superior entitlement.

2. **Legitimate occupancy and dependence.** Persons and communities acquire strong claims where their lives, institutions, identities, and continued existence genuinely depend upon resources.
3. **Creation strengthens ownership.** Creating, improving, and investing generates strong claims—but contribution does not justify unlimited control over underlying uncreated scarcity.
4. **Positive sufficiency.** Every established form of agency has a presumptive claim to enough resources for non-dominated continuation.
5. **No unlimited expansion right.** Continuation does not entail a right to endless reproduction or territorial growth.
6. **Reproductive responsibility.** Creators of new dependent persons bear special responsibilities toward them; creation cannot manufacture prior resource claims.
7. **Identity invariance.** Forking and merging do not mechanically multiply or erase resource rights; claims follow morally relevant interests and dependencies.
8. **Hibernation changes timing, not moral worth.** Suspension can reduce scarcity's cost, but active historical participation cannot be systematically monopolized by another class.
9. **Anti-preemption.** First arrival does not justify permanent enclosure of unused natural possibilities.
10. **Non-domination.** Resource institutions must not place one form of agency under another's arbitrary material control.
11. **Non-circumvention.** A community's resource and territorial claims cannot be dissolved by engineering its gradual disappearance—through manipulation, assimilation, or imposed attrition—any more than by direct dispossession.
12. **Protection of irreplaceable agency.** Unique forms of consciousness and civilization deserve strong protection against involuntary irreversible elimination.
13. **Future option space.** Present generations should not unnecessarily close every avenue through which later persons could become independent agents.
14. **Self-preservation under tragic scarcity.** Where a genuine kill-or-die conflict survives every alternative, each party may defend its continued existence; impartial procedures remain preferable where accepted, and victory establishes power, not entitlement.

15. Applications to the Present World

The theory is designed for artificial persons, hibernators, and merged minds, but if genuinely universal it should already speak to a world with none of these—for the structural problems exist in simpler forms.

15.1 Sovereignty, territory, and windfalls

Present institutions combine territorial sovereignty, private ownership, concession, inheritance, and state control: a resource within a border falls under that state's jurisdiction. This is practical, but a border tells us who governs a place, not why present citizens possess unlimited entitlement to every uncreated scarcity within it. The theory preserves sovereignty while denying that sovereignty alone settles moral ownership. It also *strengthens* some territorial claims: communities have powerful rights against displacement because homes, cultures, livelihoods, and institutions depend on place, and Indigenous communities may have especially strong claims

where colonial institutions destroyed those relationships. But it distinguishes such claims from unlimited ownership of geological windfalls. A community may have an extremely strong right to decide whether a destructive mine operates beneath its territory—the risks fall on it, its autonomy matters—without every dollar of the mineral's scarcity rent resting on the same foundation. Territorial protection grows stronger; natural-resource windfalls grow less absolute.

15.2 Scarcity rents

Institutions often let private or national actors capture enormous value merely because a naturally scarce opportunity falls within their control. Innovation, exploration, infrastructure, and risk deserve reward; but naturally occurring scarcity does not become morally identical to created value because someone holds legal title. This supports stronger public capture of portions of land value, mineral, oil, spectrum, and water rents—an old Georgist idea (George 1879), structurally anticipated by left-libertarian proposals to charge equalization fees for resource use and distribute the proceeds through a common fund (Steiner 1994; Otsuka 2003). A mining company may deserve large profits for discovery and engineering; the claim that weakens is "because we control the deposit, the entire scarcity value is ours indefinitely."

15.3 Priority without eternal ownership

Priority deserves reward, but reward is not title. Imagine a consortium that opens a remote extraction province two decades before any rival could follow: it merits secure development rights, guaranteed planning horizons, and returns large enough to recoup its extraordinary costs several times over—say, a century of exclusive operation. None of this purchases perpetual exclusion from the underlying opportunity. A twenty-year head start justifies substantial advantage; it does not justify eternity, because being somewhat faster than the next actor is a fact about timing, not a permanent moral achievement.

15.4 Depletion and later generations

When a society consumes a nonrenewable resource, later generations cannot consume it again. Depletion is not always wrong—resources can become infrastructure, knowledge, and productive capacity that leave future persons better off. The proper question is not "was it consumed?" but "was the natural opportunity transformed so as to leave adequate future possibility?" Economics names the prescriptive core: Hartwick's rule, investing exhaustible-resource rents in reproducible capital so consumption need not decline across generations (Hartwick 1977). The moral generalization: substantial portions of such rents should become durable public or intergenerational assets, not ordinary current income. A society extracting oil is converting one form of inherited possibility into another.

15.5 Environmental protection as option space

The theory does not require preserving nature untouched; it directs attention to *irreversibility*. Destroying a unique ecosystem may remove possibilities that cannot be recreated; extracting a mineral is less problematic where equivalent capabilities are preserved. Development is legitimate where it expands or maintains future capabilities; irreversible destruction requires much stronger justification.

15.6 Present-day non-domination

A nation may be sovereign yet dangerously dependent on another actor for energy, food, critical minerals, communications, finance, or essential technology; a population may hold legal rights while access to necessities is concentrated in few hands. The theory does not demand self-sufficiency, but it gives moral reason to resist extreme chokepoint control, supporting redundancy, diversification, shared governance, and limits on monopolistic control of essentials.

15.7 Inheritance

There are strong reasons to let families transfer created wealth: people plan across generations, and businesses, homes, and savings embody legitimate projects. But claims to uncreated natural scarcity have a weaker intergenerational foundation. Transmitting a built factory preserves created capital; inheriting an enormous mineral rent because an ancestor controlled the right land is different. The theory supports distinguishing more sharply between *created capital* and *inherited scarcity rents*—not abolishing inheritance.

16. Objections and Replies

16.1 The theory is premature

There are no recognized artificial persons today; why design resource morality around hypothetical minds? Because the thought experiment reveals weaknesses that already exist. Forking exposes the failure of headcount, and human population growth already strains per-capita allocation. Merging exposes the instability of numerical identity, and corporations, states, and communities already show that collective claims do not map onto individual numbers. Hibernation exposes temporal allocation, and environmental policy already distributes burdens across generations. Artificial expansion exposes permanent first possession, and colonial history already demonstrated that danger. AGI does not create these problems; it removes the biological constraints that let us avoid seeing them clearly.

16.2 Existing theories can absorb the cases

Left-libertarianism already reconciles strong property with equal claims to natural resources (Steiner 1994; Otsuka 2003); why not simply strengthen the Lockean proviso? Because these theories remain indexed to persons. An equal-share principle with per-capita compensation is exactly what the forking scenario multiplies—create a thousand copies, collect a thousand shares—and an egalitarian proviso is hostage to whatever population mechanics the future permits. The failure lies not in the egalitarian reading of acquisition but in the person-counting metric shared by all its variants. Claims must be counted against morally significant interests and dependencies, not heads: a different theory, though one that happily inherits the left-libertarian diagnosis of uncreated scarcity and its proposal to capture scarcity rents publicly.

16.3 Artificial persons are too speculative to bear rights

For the range of positions in this debate, see Schwitzgebel and Garza (2015) and Gunkel (2018). Three replies. First, the theory is conditionally stated: its principles govern whatever persons exist, and every one of them does work in the human-only present (§15). Second, the conditional is not remote: whether artificial persons with moral standing could exist is a live question, and

design decisions now already shape whether such beings will exist and on what terms (Schwitzgebel and Garza 2015). A theory of resource justice that cannot in principle accommodate non-biological persons risks falsification by its own future. Third, even if artificial persons never arrive, the diagnosis stands: our conventions rest on biological contingencies, and seeing this improves the theory we apply to ourselves.

16.4 Sufficiency and non-domination are too indeterminate

"Enough for non-dominated continuation" is said to be too vague to guide institutions, compared with crisp metrics like equal shares or first possession. But the crisp metrics are the ones that collapse, and the proposed standards yield determinate verdicts in the cases that matter. The domination test asks a concrete question—can another actor, by unilateral choice, decide whether this form of agency continues to exist? A small lunar settlement mining its own ice and reactor fuel is not dominated; a prosperous colony whose air, power, and uplink all lease from one orbital landlord is dominated. Sufficiency is fixed by the conditions of continuing agency rather than a welfare threshold—scale-relative, but not arbitrary: what a biological population needs to continue as biological differs from what a computational civilization needs, and evaluating trajectories over time rather than snapshots supplies the structure for comparison. Where indeterminacy remains, it is the indeterminacy of the subject matter; rival theories do not remove it, they hide it in false precision.

17. Conclusion

A mature theory of natural resources cannot begin and end with *who got here first*—nor rest on who worked hardest, has the most descendants, is smartest or most productive, holds legal title, or replicates fastest. These rules become unstable once persons can exist in radically different forms.

A universal theory must instead protect the conditions under which different forms of agency coexist without domination. It must recognize legitimate occupancy without turning historical presence into eternal ownership; reward creation without confusing created value with ownership of nature; allow reproduction without letting it become territorial conquest; survive forking and merging; recognize hibernation without making dormant persons permanent outsiders; protect irreplaceable forms of consciousness without freezing all change; preserve future possibility without pretending future persons already own the present; and admit that under tragic scarcity morality may permit self-preservation without pretending the victor was morally superior.

The deepest principle is not equal ownership of the universe but **equal standing within the universe**. A civilization holding a thousandth of the galaxy's matter may be a full moral equal of one holding half of it, provided each answers to no other's arbitrary will. What would make it unjust is not the percentage, but existence at another's discretionary permission, the permanent enclosure of every avenue of future independence, or the conversion of temporary advantage into eternal entitlement:

> **Finite reality is a shared field of possibility through which successive persons construct lives. Existing persons may acquire powerful rights of occupancy, use, creation, reliance, and control. But no species, generation, lineage, or architecture**

of mind gains an unlimited right to turn historical priority into permanent domination over all who may later exist.

Property remains necessary; markets remain possible; sovereignty remains useful; inheritance remains legitimate in many forms. What disappears is the assumption that ownership of uncreated nature is naturally absolute and eternally settled. We are not the final owners of the material future, because we are not the final inhabitants of it.

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